

Introduction to probability

Experiment:- An experiment is virtually any operation that results in 1 or more outcomes.

for ex:- Appearing for F.Y. Bsc examination is an experiment with possible outcomes as pass or fail.

The are 2 types of experiment :-

1. Deterministic Experiment:- A deterministic

experiment is an experiment for which the outcome is unique hence certain. The outcome can therefore be predicted before performing the experiment.

for example:- Cooling water below zero degree celcius = It will freeze

2. Non-deterministic Experiment / Random experiment

In this experiment there are more than one possible outcomes and the result of experiment cannot be predicted in advance
for example:- Tossing of a coins = head/tail

* Sample space:- The set of all possible distinct outcomes of an experiment is called a sample space.

For example:- 1) Tossing a coin

$$S = \{H, T\}$$

2) Rolling a die

$$S = \{1, 2, 3, 4, 5, 6\}$$

There are 2 types of a sample space

- i) Discrete Sample space
- ii) Continuous Sample space

1) Discrete Sample Space:- A sample space containing a finite number of points or countably infinite points is called a discrete sample space.

There are 2 types of discrete sample space:-

i) Finite Sample space:- If the number of elements contain in the sample space is finite then it is called as finite sample space.

ex:- Suppose 2 coins are tossed
 $S = \{HH, HT, TH, TT\}$

possible

Countably infinite sample space:- If the number of elements in Ω are countably infinite such a sample space is called countably infinite sample space.

ex:- Number of accidents on Mumbai-pune highway in a month.

$$\Omega = \{0, 1, 2, \dots\}$$

ii) continuous sample space:- A sample space

Ω is called a continuous sample space if the number of elements in Ω are uncountably infinite.

ex:- Consider the experiment of measuring height of a person. we may take

$$\Omega = \{0, \infty\}$$

Events:- Any subset of a sample space is called as event.

for eg:- Rolling a die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

It consists of sum of all points of the sample space

for example:- Suppose the experiment of rolling a die

• Sample space
• Events

Date: / /

$\Omega = \{1, 2, 3, 4, 5, 6\}$

Let A be the event that even number appears on upper most face of the die

$A = \{2, 4, 6\}$

• Types of events

[1] Simple event / Elementary events: An

event containing only one outcome is called as simple event.

For example:

$A =$ getting a multiple of 5 on a die

$A = \{5\}$

[2] Impossible event:

An event corresponding to empty set is called as impossible event. For example: If a single coin is tossed then getting 2 heads is an impossible event.

[3] Sure event / certain event :-

An event containing all the points of Ω is called a sure event. For example: Getting a number either even or odd on a rolled die is a

[4] Multi

[5]

[6]

sure event:

[4] Mutually Exclusive Event:- (Disjoint event) :-

Event A and B are said to be mutually exclusive event if there is no common elements in A and B. i.e. $A \cap B = \phi$

ex:- $\omega = \{ \}$ Throwing a die

$$S = \{1, 2, 3, 4, 5, 6\}$$

A be the event an even number occurs

$$A = \{2, 4, 6\}$$

B be the event such that odd number occurs

$$B = \{1, 3, 5\}$$

$$\text{Therefore, } A \cap B = \{ \} = \phi$$

[5] Exhaustive Event:-

Events A and B are said to be exhaustive event if $A \cup B = \omega$

$$\text{ex:- } \omega = \{1, 2, 4, 3, 5, 6\}$$

$$A_1 = \{1, 3\}$$

$$A_2 = \{2, 3, 5\}$$

$$A_3 = \{4, 5, 6\}$$

$$\therefore A_1 \cup A_2 \cup A_3 = \{1, 2, 3, 4, 5, 6\} = \omega$$

[6] Complement of an event:-

A-complement is an event containing all points in ω which are not in A. A^c

ex:- $\Omega =$ Two coins tossed simultaneously
 $\Omega = \{HH, HT, TH, TT\}$

A be the event that 2 head occurs.

$$A = \{HH\}$$

$$A^c = \{HT, TH, TT\}$$

7) Occurrence of atleast one of the given events

If A and B are 2 events on Ω , then occurrence of atleast one of the events A and B is defined as $A \cup B$.

$$\text{ex:- } \Omega = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

A: occurrence of even no.

$$A = \{2, 4, 6, 8\}$$

B: occurrence of multiple of 3.

$$B = \{3, 6\}$$

here occurrence of number either even or multiple of 3 is given by
 $A \cup B = \{2, 4, 6, 8, 3, 6\}$

8) Occurrence of all the given events/simultaneous occurrence of the events:-

If A and B are events on Ω , then $A \cap B$ is the occurrence of both the events A & B. It is also called as simultaneous occurrence of events A & B.

eg:- $\Omega =$

$A: 1, 2$

$A = \{1, 2\}$

$B: 2, 3$

$B = \{2, 3\}$

$\therefore A \cap B$

we

if

13) Occ

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egs

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then

events

given events.

then

events

even

alternates

$A \cap B$

$A \& B$

occurrence

eg:- $\Omega = \{1, 2, 3, 4, 5, 6\}$

A: be the event that multiple of 3.

$$A = \{3, 6\}$$

B: occurrence of an even no.

$$B = \{2, 4, 6\}$$

$$\therefore A \cap B = \{6\}$$

we say that A and B occur simultaneously if 6 appears.

9] Occurrence of non. of the given events:-

Occurrence of non of the events A & B is

given by $(A \cup B)^c = A^c \cap B^c$

$$\text{egs:- } \Omega = \{1, 2, 3, 4, 5, 6\}$$

A: even no. appears

$$A = \{2, 4, 6\}$$

B: multiple of 3 appears

$$B = \{3, 6\}$$

Then non of the events A and B appear is

$$(A \cup B)^c = \{2, 3, 4, 6\}^c = \{1, 5\}$$

$$A^c \cap B^c = \{1, 5\}$$

10] Relative complement: The relative complement

of A with respect to B is given by

$$A^c \cap B \text{ i.e. the set of points which}$$

are not in A but in B.

similarly, the relative complement of B with respect to A is given by $A \setminus B$.

Ex 1] Write down sample spaces for the following experiments. Also, state the type of the sample space.

(i) Examination results (P: pass, F: fail) are noted for 3 students

(ii) 10 radio sets are checked and number of defective sets are noted.

(iii) Items ~~turning off~~ a production ~~light~~ ~~are~~ ~~marked~~ ~~defect~~

(iv) A coin is tossed until head appears for the first time.

(v) $\Omega = \{PPP, PFP, PFF, FPP, FPF, FFF\}$
 $n(\Omega) = 8$

Such a sample space is called finite sample space

(vi) $\Omega = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
 $n(\Omega) = 11$

Such a sample space is called finite sample space

Note:-

iii) $\Omega = \{H, TH, TTH, TTTH, \dots\}$

This type of sample space is called continuous

sample space

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Ex 2) Let A, B, C be any three events on a sample space. Write expressions for the events

(a) At least one of the events A, B, C occurs

(b) Only A occurs

(c) A and B occurs but not C.

(d) All three events occurs

(e) None of A, B, C occurs

(f) Exactly one occurs

(g) Exactly two occurs.

$\Rightarrow$ (a) $= A \cup B \cup C$

(b) $= A \cap B' \cap C'$

(c) $= A \cap B \cap C'$

(d) $= A \cap B \cap C$

(e) $= (A \cup B \cup C)'$

(f) $= (A \cap B \cap C') \cup (A' \cap B \cap C) \cup (A \cap B' \cap C)$

(g) $= (A \cap B \cap C') \cup (A \cap B' \cap C) \cup (A' \cap B \cap C)$

2) $A \cap B \rightarrow$ all event occur / sim. occur

3) $A \cup B \rightarrow$ at least one occur

3) None of the event occur

$\rightarrow (A \cup B)'$

note:-

[for any event not coming is written as complement]

None of $\Rightarrow (A \cup B)'$

3) Ex:- consider the following sample space

$$S = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

write down the following events

(i) A: An odd number appears

(ii) B: Number is greater than 4

(iii) $A \cup B$

(iv) both A and B occur

(v) None of the A & B occurs

(vi) Relative complement of A with respect to B occurs

(vii) A occurs but not B

Ans $\Rightarrow$ (i) $A = \{1, 3, 5, 7\}$

$$(ii) B = \{5, 6, 7, 8\}$$

$$(iii) A \cup B = \{1, 3, 5, 7, 6, 8\}$$

$$(iv) A \cap B = \{5, 7\}$$

$$(v) ~~A \cap B~~ (A \cup B)^c = \{2, 4\}$$

$$(vi) A' \cap B \Rightarrow$$

$$A' = \{2, 4, 6, 8\}$$

$$A' \cap B = \{6, 8\}$$

(vii) $A \cap B'$

$$B' = \{1, 2, 3, 4\}$$

$$A \cap B' = \{1, 3\}$$

* Probability of an event :- Let A be the any event of sample space S . then probability of A is

$$A = \frac{\text{no. of elements belonging to } A}{\text{Total no. of elements in } S} = \frac{n(A)}{n(S)} = \frac{m}{n}$$

Note that, $0 \leq P(A) \leq 1$

Ex] If a pair of unbiased coins is tossed, obtain probability of occurrence of (i) both heads, (ii) single heads, (iii) Atleast one head
 $\Rightarrow$ Sample space: A pair of unbiased coins is tossed

$$S = \{HH, HT, TH, TT\}$$

$$n(S) = 4 = n$$

(i) A : both head appear

$$A = \{HH\}$$

$$n(A) = 1 = m$$

$$P(A) = \frac{m}{n} = \frac{1}{4}$$

(iv) B: Single head appears

$$B = \{TH, HT\}$$

$$n(B) = 2 = n$$

$$P(B) = \frac{n}{n} = \frac{2}{4} = \boxed{\frac{1}{2}} = 0.5$$

(v) C: At least one head

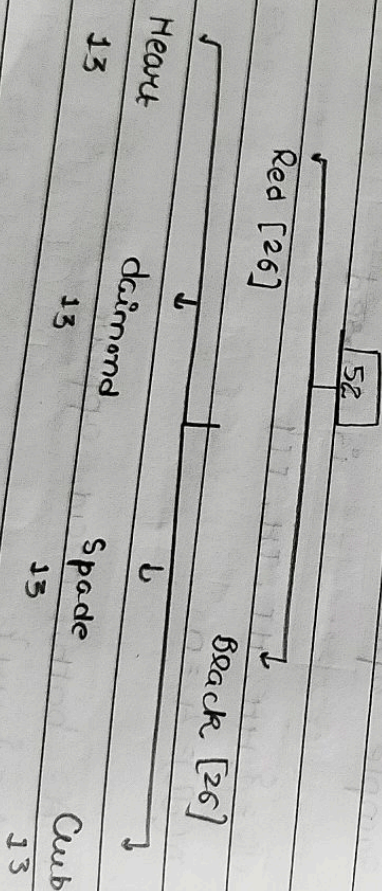
$$C = \{HT, TH, HH\}$$

$$n(C) = 3 = n$$

$$P(C) = \frac{n}{n} = \frac{3}{4} = \boxed{\frac{3}{4}} = 0.75$$

Ex 2) 4 cards are drawn at random from a well shuffled pack of 52 cards find the probability that:-

- (1) 2 cards are red and 2 are black
- (2) All cards are of different suits
- (3) All are of same suit
- (4) one is king



→ n = 5

(i) A: Two

A =

n(A) =

P(A)

(ii) B:

A =

=

=

=

P(A)

$$\rightarrow n = {}^{52}C_4 = \frac{52!}{4!} = \frac{52 \times 51 \times 50 \times 49 \times 48!}{4 \times 3 \times 2 \times 1 \times 48!}$$

$$n(CD) = 270,725$$

(1) A: Two cards are red & 2 are black

$$A = {}^{26}C_2 \times {}^{26}C_2$$

$$= \frac{26 \times 25 \times 24!}{24! \times 2!} \times \frac{26 \times 25 \times 24!}{24! \times 2!}$$

$$= [325] [325]$$

$$n(A) = 105,625$$

$$P(A) = \frac{105,625}{270,725} = \boxed{0.3901}$$

(ii) B: All cards are of different suits

$$B = {}^{13}C_1 + {}^{13}C_1 + {}^{13}C_1 + {}^{13}C_1$$

$$= 13! \times 4 \times \frac{13!}{1!} \times 4 \times \frac{13!}{1!} \times 4 \times \frac{13!}{1!}$$

$$= \frac{13 \times 12! \times 4 \times 13 \times 12! \times 4 \times 13 \times 12! \times 4 \times 13 \times 12!}{12! \times 12! \times 12! \times 12!}$$

$$= 13 \times 13 \times 13 \times 13$$

$$= 28561$$

$$P(B) = \frac{28561}{270725} = \boxed{0.1054}$$

(iii) All are of same suit:-

C: All are of same suit

$$n(C) = {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4$$

$$= \left[\frac{13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{91 \times 4 \times 3 \times 2 \times 1} \right] \times 4$$

$$n(C) = [26 \times 11 \times 10]$$

$$P(C) = \frac{n(C)}{n(S)} = \frac{26 \times 11 \times 10}{270725} = \frac{2860}{270725} = 0.01056$$

(iv) One is King:-

D: One card is a King card.

$$n(D) = {}^4C_1 + {}^{48}C_3$$

$$= \frac{4 \times 3! \times 1}{3! \times 1} \times \frac{24 \times 8}{48 \times 47 \times 46 \times 45 \times 44 \times 43 \times 42 \times 41}$$

$$= 4 \times (8 \times 47 \times 46)$$

$$n(D) = 12976$$

$$= 11300$$

$$P(D) = \frac{n(D)}{n(S)} = \frac{12976}{270725} = 0.0479$$

Ans

$$P(A) = 0.3901$$

$$P(B) = 0.1054$$

$$P(C) = 0.01056$$

$$P(D) = 0.0479$$

Continue

$$n(D) = (17, 296) (4)$$

$$n(D) = 69, 184$$

$$P(D) = \frac{69184}{270725} = 0.2555$$

Probability Model:-

Let Ω be a finite sample space containing value $\omega_1, \omega_2, \dots, \omega_n$ then $P(\omega)$ is said to be probability model if it satisfies the following conditions:-

- (i) $P(\omega_i) \geq 0 \quad \forall i$
- (ii) $\sum_{i=1}^n P(\omega_i) = 1$

~~or~~ i.e. $P(\omega_1) + P(\omega_2) + P(\omega_3) + \dots + P(\omega_n) = 1$

Ex: i) Suppose $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6\}$

(i) $\Omega_1 = \{0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}\}$

(ii) $\Omega_2 = \{\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\}$

(iii) $\Omega_3 = \{\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\}$

(iv) $\Omega_4 = \{-\frac{1}{3}, \frac{2}{3}, 0, \frac{1}{3}, 0, 0\}$

which of the following probability models are verified.

Ans $\rightarrow$ (i) $\Omega_1 = \{0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}\}$

~~$P(\Omega_1) =$~~

$P(\omega_1) = 0 \geq 0$

$P(\omega_2) = 0 \geq 0$

$P(\omega_3) = 0 \geq 0$

$P(\omega_4) = 0 \geq 0$

$P(\omega_5) = \frac{1}{2} \geq 0$

$P(\omega_6) = \frac{1}{2} \geq 0$

$$\frac{1}{2} \sum_{i=1}^3 P(\omega_i) = 0 + 0 + 0 + 0 + 1 + 1$$

$$\sum_i p(\omega_i) = 1$$

∴ It is a probability model

③

(ii) $\mathcal{A}_2 = \{\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\}$

1) $P(\omega_1) = \frac{1}{4} \geq 0$

$P(\omega_2) = \frac{1}{4} \geq 0$

$P(\omega_3) = \frac{1}{4} \geq 0$

$P(\omega_4) = \frac{1}{4} \geq 0$

$P(\omega_5) = \frac{1}{4} \geq 0$

$P(\omega_6) = \frac{1}{4} \geq 0$

2) $\sum_{i=1}^6 P(\omega_i) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

$= \frac{6}{4}$

$= \frac{3}{2} = \boxed{1.5}$

$\therefore$ It is not a probability model

$\mathcal{A}_3 = \{\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}\}$

17 $P(\omega) = \frac{1}{4} \geq 0$

$$P(\omega_2) = 1/4 \geq 0$$

$$P(\omega_3) = 1/4 \geq 0$$

$$P(\omega_4) = 1/4 \geq 0$$

$$P(w_5) = 1/4 \geq 0$$

$$P(\omega_0) = 1/4 \geq 0$$

$$c_w p(w_i) = 1 + 1 + 1 + \dots$$

[illegible]

4/5

[illegible]

$$P(w_i) \neq 1$$

∴ It is not a

(iii)

$$Q_3 = \{ \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1 \}$$

$$P(u_i) = \frac{1}{8}$$

$$P(\omega_2) = 1/8$$

$$P(w_3) = \frac{1}{4}$$

$$P(w_4) = \frac{1}{8}$$

$$P(w_5) = 0$$

$$P(w_6) = \frac{1}{8}$$

$$\sum_{i=1}^n P(w_i) = \frac{1}{8} + \frac{1}{8} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8}$$

$$= \frac{4+1}{8}$$

$$= \frac{4+2}{8}$$

$$= \frac{6}{8}$$

$$= \frac{3}{4} = \boxed{0.75}$$

$\therefore \sum_{i=1}^n P(w_i) \neq 1$ but equal to 0.7 thus it

does not satisfy 2nd condition. Therefore it is not a probability model.

(iv) (i) $P(w_i) \geq 0$

but $P(w_i) = -\frac{1}{3} \neq 0$

Thus, it is not a probability model.

Since 1st condition is not satisfied thus
It is not a probability model.

Ex 2] A loaded die has following probability
assignment to the six faces.

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$\text{with } P(1) = 0.1$$

$$P(2) = 0.2$$

$$P(3) = 0.3$$

$$P(4) = 0.25$$

$$P(5) = 0.1$$

$$P(6) = 0.05$$

What is the probability of

(i) An even number appears

(ii) A number less than 4 appears.

(i) Let A be an event having an even
number appears

$$A = \{2, 4, 6\}$$

$$P(A) = P(2) + P(4) + P(6)$$

$$= 0.2 + 0.25 + 0.05$$

$$P(A) = 0.5$$

(ii) Let B be an event having number
less than 4 appears

$$B = \{1, 2, 3\}$$

$$P(B) = P(1) + P(2) + P(3)$$

$$= 0.1 + 0.2 + 0.3$$

$$P(B) = 0.6$$

Ans $\Rightarrow P(A) = 0.5$ and $P(B) = 0.6$

Axioms of probability :-

Let Ω be a sample space and A be the any event of Ω then $P(A)$ is the probability of an event A which satisfies the following axioms:-

Axiom (1) $P(A) \geq 0$

Axiom (2) $P(\Omega) = 1$

Axiom (3)

Let $A_1, A_2, \dots, A_n$ are any mutually exclusive event

then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

Imparticular IF A & B are any two mutually exclusive event then $P(A \cup B) = P(A) + P(B)$

Theorem (1) :-

$$P(A^c) = 1 - P(A)$$

prove it:-

$$\Rightarrow A^c \cup A = \Omega$$

$$\therefore P(A^c) + P(A) = P(\Omega)$$

$$\therefore P(A^c) + P(A) = 1$$

$$\text{---} (\because P(\Omega) = 1)$$

$$\therefore P(A^c) = 1 - P(A)$$

Hence proved.

Theorem ② :- $P(\phi) = 0$ prove " "

$$\phi^c = \Omega$$

$$P(\phi^c) = P(\Omega)$$

$$\therefore 1 - P(\phi) = P(\Omega)$$

$$1 = P(\phi) + 1$$

$$\therefore P(\phi) = 0$$

Hence proved.

Theorem ③ :- $0 \leq P(A) \leq 1$

→ **Axiom ①** $P(A) \geq 0$ ——— (1)

$$P(A^c) \geq 0 \text{ ——— (2)}$$

$$1 - P(A) \geq 0$$

$$P(A) \leq 1 \text{ ——— (3)}$$

From (1) and (3),

$$0 \leq P(A) \leq 1$$

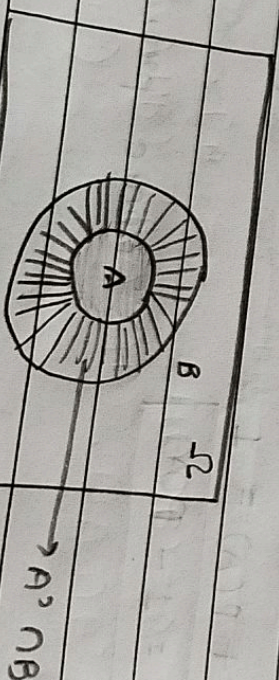
Hence proved

Theorem ④ :-

→ If $A \subset B$ then $P(A) \leq P(B)$

$$P(B) = P(A) + P(A^c \cap B) \geq 0$$

$$P(B) \geq P(A)$$



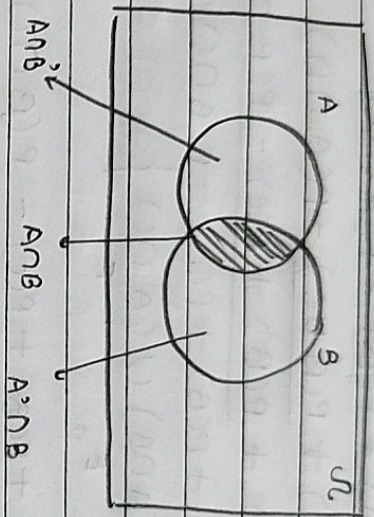
Theorem

Theorem (5) :-

Addition theorem of probability
Let A & B are 2 events of Ω then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Ans $\Rightarrow$



$$A \cup B = (A \cap B') + (A \cap B) + (A' \cap B)$$

$$P(A \cup B) = P(A \cap B') + P(A \cap B) + P(A' \cap B) \quad \text{--- (1)}$$

$$P(A) = P(A \cap B') + P(A \cap B) \quad \text{--- (2)}$$

$$P(B) = P(A \cap B) + P(A' \cap B) \quad \text{--- (3)}$$

$$P(A) + P(B) = P(A \cap B') + P(A \cap B) + P(A' \cap B) + P(A \cap B)$$
$$P(A \cap B)$$

$$= P(A \cup B) + P(A \cap B)$$

$$P(A) + P(B) - P(A \cap B) = P(A \cup B)$$

12/12/24

Theorem 6 :-

If A, B and C are any three events
then $P(A \cup B \cup C) = P(A) + P(B) + P(C) -$

$$P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

→ $KHS = P(A \cup B \cup C)$

consider $(B \cup C) = D$

$$= P(A \cup D)$$

$$= P(A) + P(D) - P(A \cap D)$$

$$= P(A) + P(B \cup C) - P(A \cap D)$$

$$= P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap B) - P(A \cap C)$$

$$= P(A) + P(B) + P(C) - P(B \cap C) -$$

$$P[(A \cap B) \cup (A \cap C)]$$

$$= P(A) + P(B) + P(C) - P(B \cap C) - P(E \cup F)$$

$$= P(A) + P(B) + P(C) - P(B \cap C) - [P(E) + P(F) - P(E \cap F)]$$

$$= P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap B) -$$

$$P(A \cap C) + P[(A \cap B) \cap (A \cap C)]$$

$$= P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$$

Hence proved

Ex 1]

$$P(A) = 0.6, P(B) = 0.5, P(A \cap B) = 0.3$$

compute:-

$$P(A'), P(A \cup B), P(A' \cap B), P(A' \cap B')$$

and

$$\text{Ans} \rightarrow [1] P(A') =$$

$$= 1 - P(A)$$

$$= 1 - 0.6$$

$$P(A') = 0.4$$

$$P(A \cap B)$$

$$[2] P(A \cup B) = P(A) + P(B) + P(A \cap B)$$

$$= 0.6 + 0.5 - 0.3$$

$$P(A \cup B) = 0.8$$

$$[3] P(A' \cap B) = P(B) - P(A \cap B)$$

$$= 0.5 - 0.3$$

$$P(A' \cap B) = 0.2$$

$$[4] P(A' \cap B') = P(A \cup B)'$$

$$= 1 - P(A \cup B)$$

$$= 1 - 0.8$$

$$P(A' \cap B') = 0.2$$

$$[5] P(A' \cup B') = P(A \cap B)'$$

$$= 1 - P(A \cap B)$$

$$= 1 - 0.3$$

$$P(A' \cup B') = 0.7$$

Ex 2: A and B are two events defined in a sample space S . Can the following statements be regarded as true? Justify your answer.

(i) $P(A) = 0.8$, $P(B) = 0.7$, $P(A \cap B) = 0.8$

(ii) $P(A) = 0.8$, $P(B) = 0.7$, $P(A \cap B) = 0.3$

Ans $\Rightarrow$ [i] 1st statement is false as $P(A \cap B) > P(B)$ which is wrong as $(A \cap B) \subset B$.
Therefore, $P(A \cap B) \leq P(B)$.

[ii] 2nd statement is also false as $P(A \cup B)$
 $= P(A) + P(B) - P(A \cap B)$
 $= 0.8 + 0.7 - 0.83$
 $= 0.8 + 0.4$
 $= 1.2 \neq 1$

* ~~Q3~~ Suppose A, B and C are three events defined in S such that
 $P(A) = P(B) = P(C) = \frac{1}{4}$

$$P(A \cap C) = \frac{1}{8}$$

$$P(A \cap B) = P(B \cap C) = 0$$

Calculate

(i) $P(A \cup B \cup C)$ (ii) $P(A \cap B \cap C)$

Ans = $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap C) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$

$$= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} - 0 - 0 - \frac{1}{8}$$

(B)

Ex) Suppose a card is drawn at random from a well shuffled pack of 52 cards. let event A =

get

$$= \frac{3}{4} - \frac{1}{8} = \frac{24-4}{32} = \frac{20}{32}$$

$$= \frac{40}{16} = \frac{5}{2} = \boxed{1.25} = \boxed{0.625}$$

(ii) $P(A \cap C) = \frac{1}{4}$

$$P(A \cup C) = P(A) + P(C) - P(A \cap C)$$

$$= \frac{1}{4} + \frac{1}{4} - \frac{1}{8}$$

$$= \frac{2-1}{4} = \frac{4-1}{8}$$

$$= \frac{3}{8} = \boxed{0.375}$$

(iii) $P(A' \cap B' \cap C') = P(A \cup B \cup C)'$

$$= 1 - P(A \cup B \cup C)$$

$$= 1 - \frac{5}{8}$$

$$= \left[\frac{3}{8} \right]$$

$$(iv) P(A \cap C) = P(A) + P(C)$$

$$P(A \cup C) = P(A) + P(C) - P(A \cap C)$$

$$P(A \cup C) - P(A) - P(C) = -P(A \cap C)$$

$$\therefore P(A \cap C) = -P(A \cup C) + P(A) + P(C)$$

$$= -$$